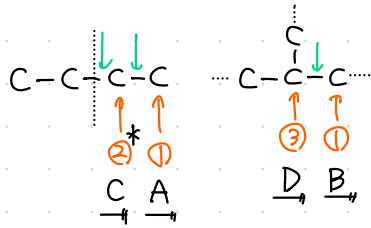
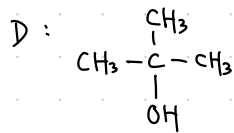
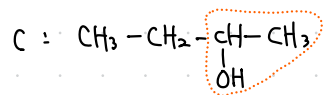
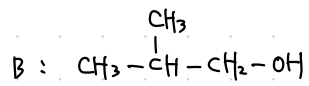
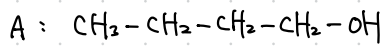


3-11 $C_4H_{10}O$ $U=1$ $\left\{ \frac{\text{アルコール}}{\text{エーテル}} \right\}$



アルコール $A^{\text{①}}$ $B^{\text{①}}$ C^* $D^{\text{②}}$
 O_x $\underline{E}$ $\underline{F}$ G X
 アルコール アルコール
 b.p. $A > B$

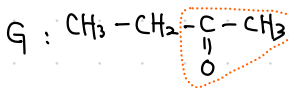
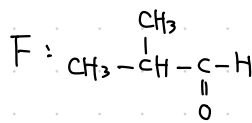
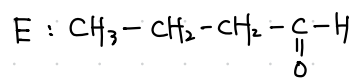
問1



問2

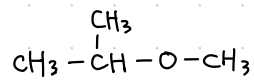
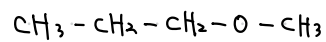
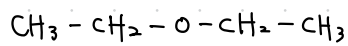
ヒドロキシ基を持ち、
分子間での水素結合を形成するため。

問3

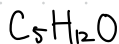


C, G

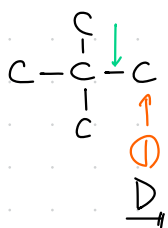
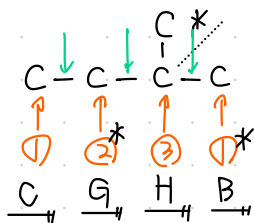
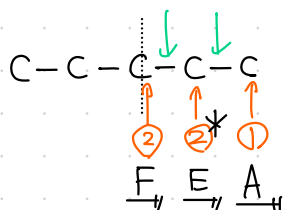
問4



3 - 2

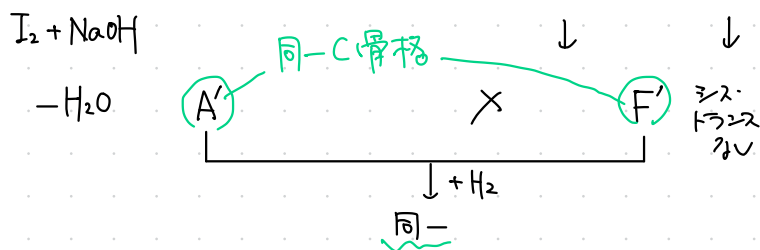


$$U=0 \quad \left\{ \begin{array}{l} \gamma_{\mu\nu} = -\gamma_{\nu\mu} \\ I = -\bar{I} \end{array} \right.$$



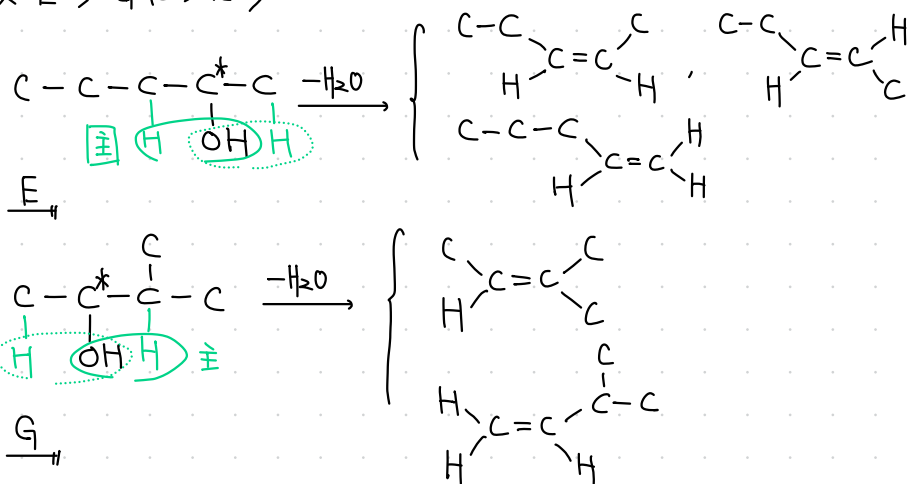
アルコール ① A ① B^{*} ① C ① D ② E^{*} ② F ③ G^{*} ③ H

Ox { アルデヒド } { ケトン } X



$\left. \begin{array}{l} \text{「脱水した} \sim \text{」} \\ \text{「H}_2\text{付加した} \sim \text{」} \end{array} \right\} \text{同一化合物}$
 $\longrightarrow \text{同一C骨格}$

$\langle 2 \text{級 } E^*, G^* | \Rightarrow WZ \rangle$

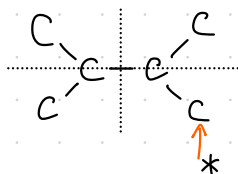
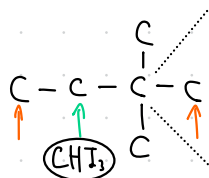
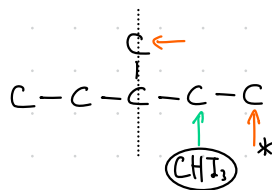
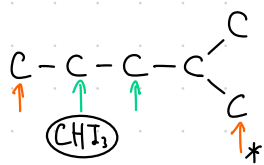
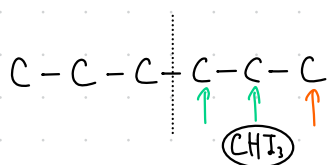


(2) $\xrightarrow{6}$ (立体异构: 7)

3 - ③

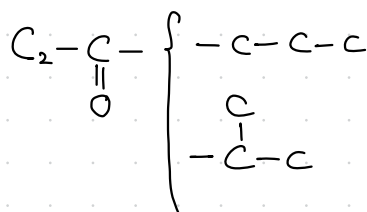
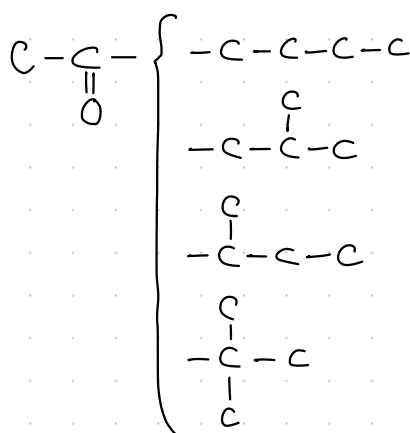
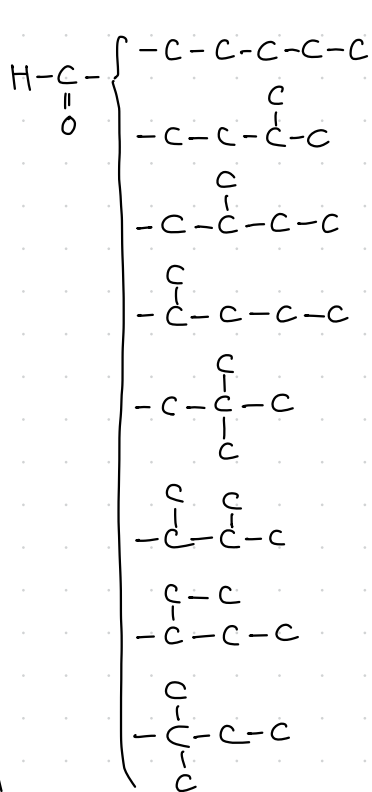
問1

← : =O (アルデヒド)
← : =O (ケトン)



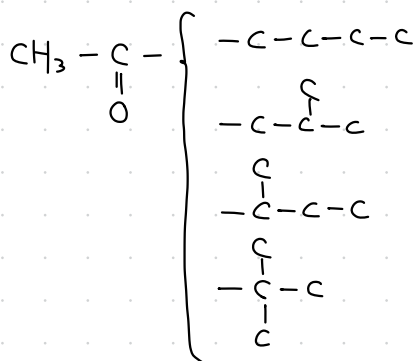
(a) 8

(b) 6

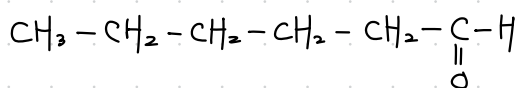


問2. 3

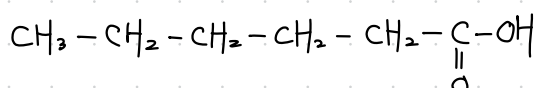
問3. 4



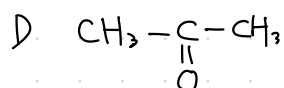
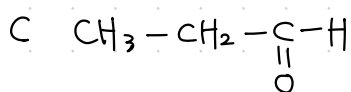
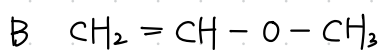
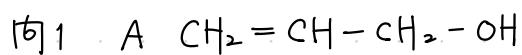
問4. X



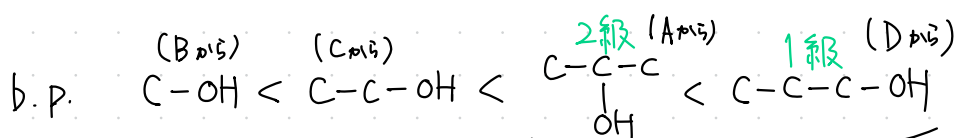
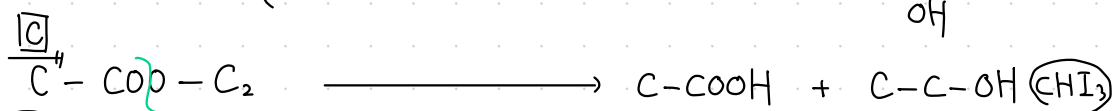
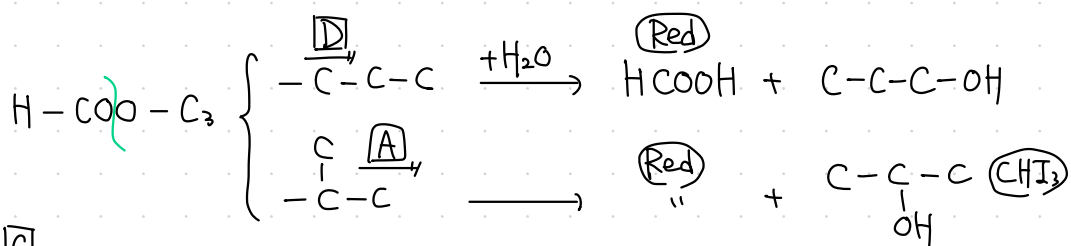
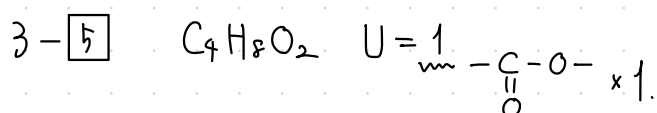
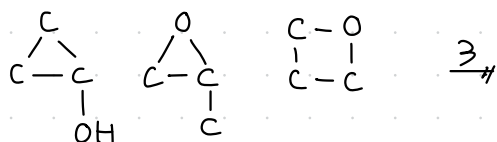
Y



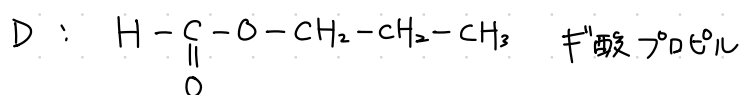
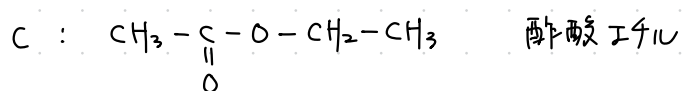
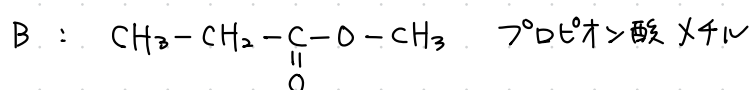
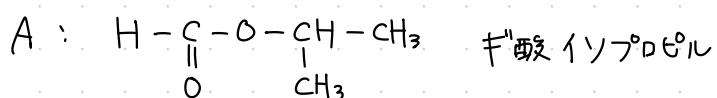
3-4



162



(小) $\leftarrow$ 分子量 $\longrightarrow$ (大) 異性体



3-6

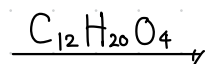
$$\text{例1} \quad w_C = 264 \text{ g} \times \frac{C(12)}{CO_2(44)} = 72 \text{ g}$$

$$w_H = 90 \text{ g} \times \frac{2H(2)}{H_2O(18)} = 10 \text{ g}$$

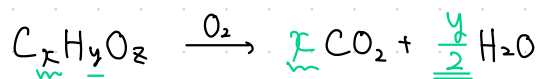
$$w_O = 114 - (72 + 10) = 32 \text{ g}$$

$$C : H : O = \frac{72 \text{ g}}{12 \text{ g/mol}} : \frac{10 \text{ g}}{1.0 \text{ g/mol}} : \frac{32 \text{ g}}{16 \text{ g/mol}} = 3 : 5 : 1$$

$$(C_3H_5O)_n = 57n = 228 \quad n = 4$$



<別解> 燃烧例, 分子量既知の時



$$C_xH_yO_z : CO_2 = 1 : x \text{ mol},$$

$$(288) \quad (44)$$

$$\frac{114 \text{ g}}{228 \text{ g/mol}} \times x = \frac{264 \text{ g}}{44 \text{ g/mol}} \times 1 \quad \therefore x = 12$$

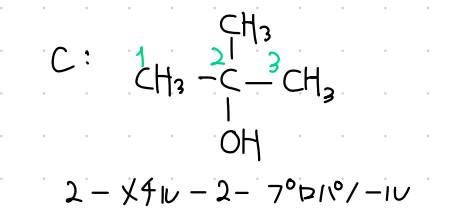
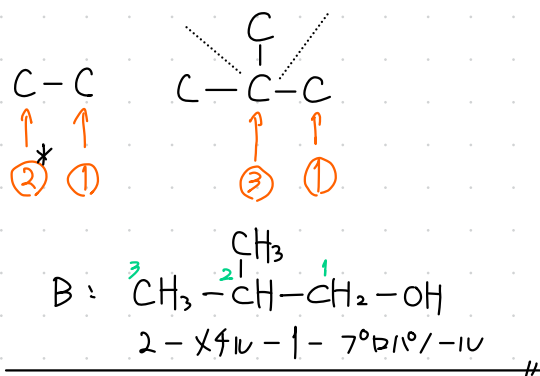
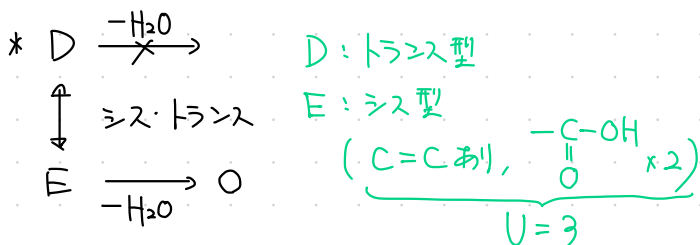
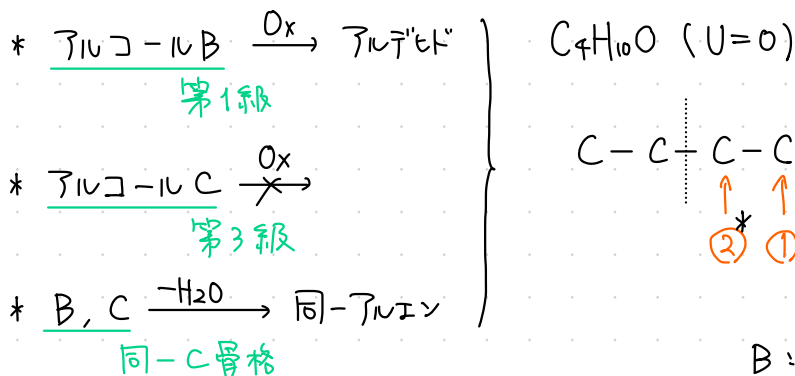
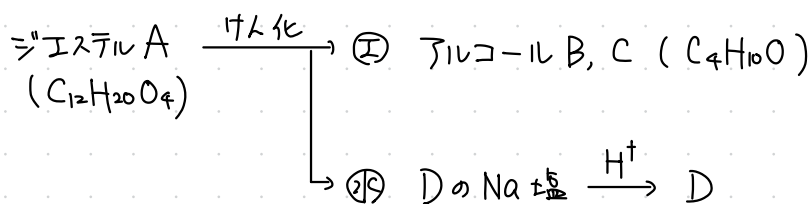
$$C_xH_yO_z : H_2O = 1 : \frac{y}{2} \text{ mol}$$

$$(288)$$

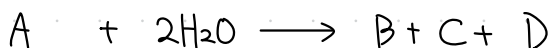
$$\frac{114 \text{ g}}{228 \text{ g/mol}} \times \frac{y}{2} = \frac{90 \text{ g}}{18 \text{ g/mol}} \times 1 \quad \therefore y = 20$$

$$C_{12}H_{20}O_z = 164 + 16z = 228 \quad \therefore z = 4 \quad \underline{C_{12}H_{20}O_4}$$

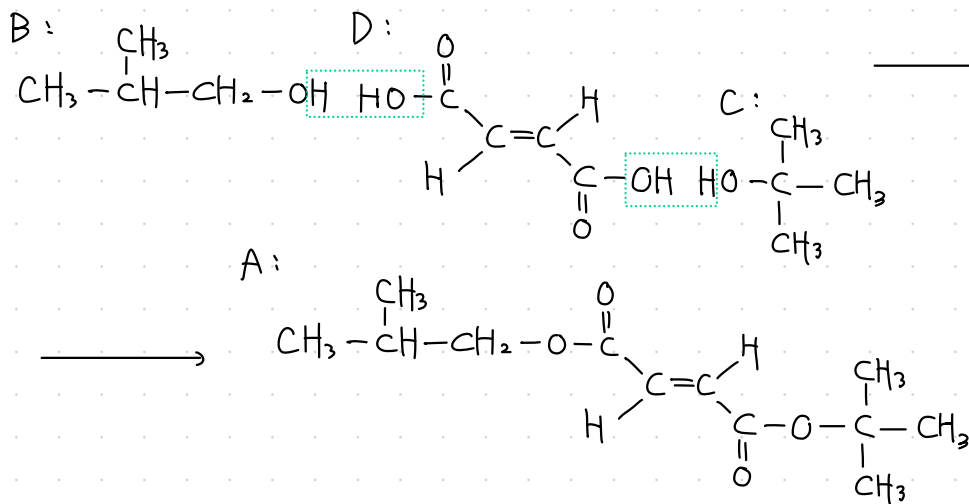
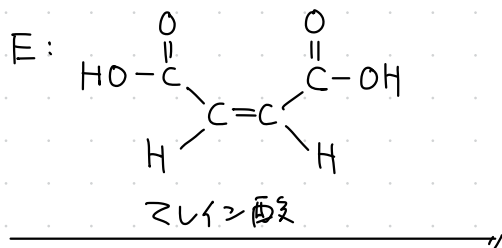
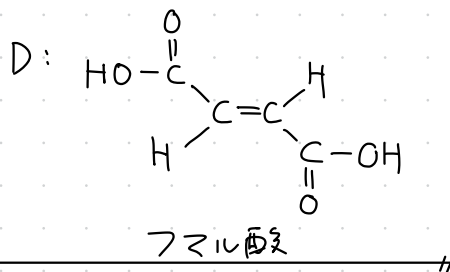
問2. A: $C_{12}H_{20}O_4$ $U = \frac{2 \times 12 + 2 - 20}{2} = 3$ C=C $\times 1$ + $-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{O}- \times 2$



ジエステル



$\therefore D = A + 2H_2O - (B + C)$
 $= C_{12}H_{20}O_4 + 2H_2O - 2 \times C_4H_{10}O$
 $= C_4H_4O_4$ (116)
 ($U=3$)



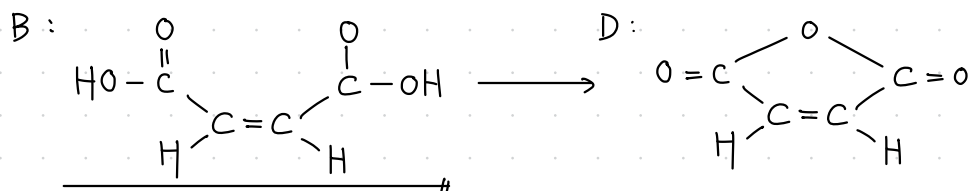
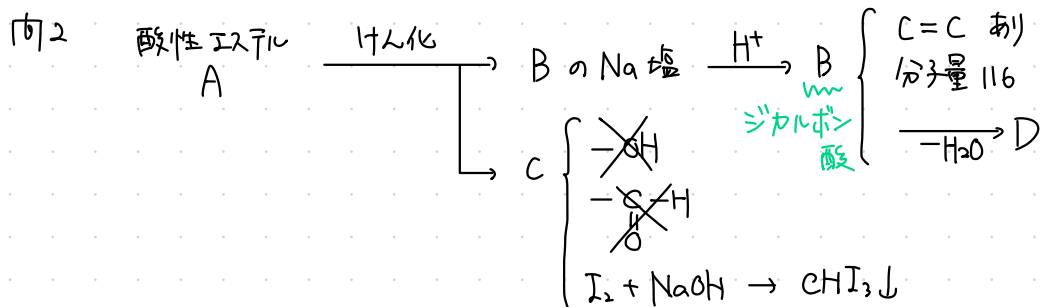
3 - [7]

問1 $C_7H_8O_4$

$$C:H:O = \frac{53.8}{12} : \frac{5.1}{1.0} : \frac{41.1}{16} = 4.48 : 5.1 : \frac{2.56}{\text{①}}$$

$$\div 1.75 : 1.99 : 1 = 7:8:4$$

$$130 \leq (C_7H_8O_4)_n \leq 170 \quad \therefore n=1 \quad \underline{C_7H_8O_4} \quad (U=4)$$

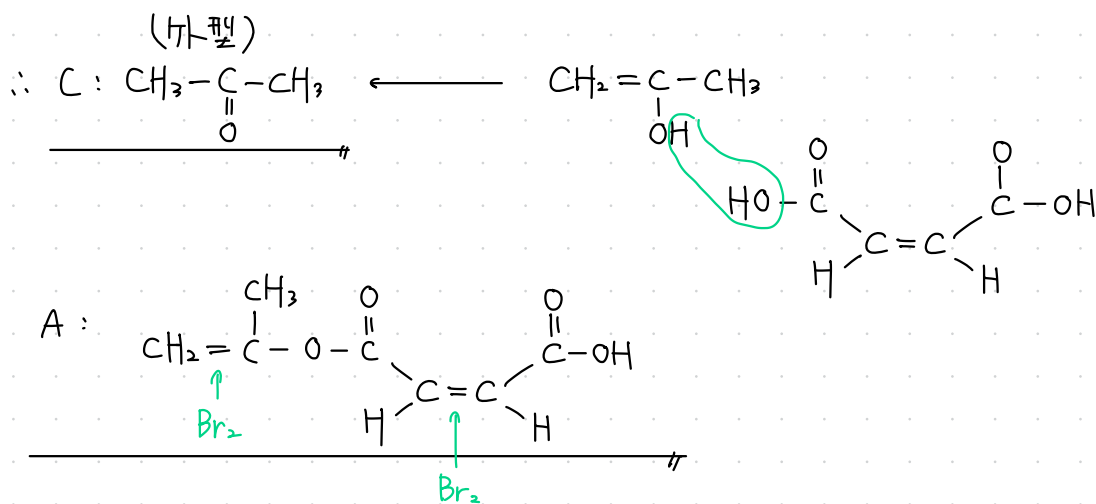


$$A + H_2O \rightarrow B + C \quad \therefore C = A + H_2O - B$$

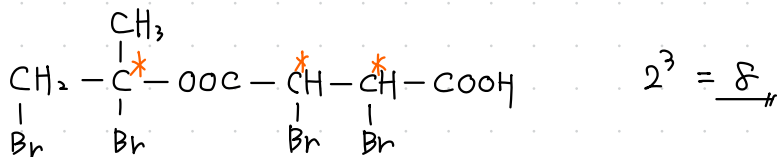
$$= C_7H_8O_4 + H_2O - C_4H_4O_4$$

$$= \underline{C_3H_6O} \quad (U=1)$$

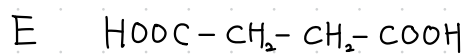
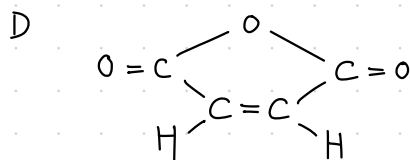
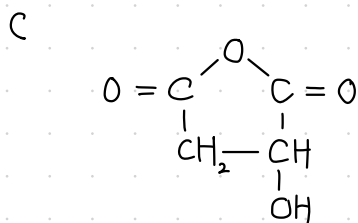
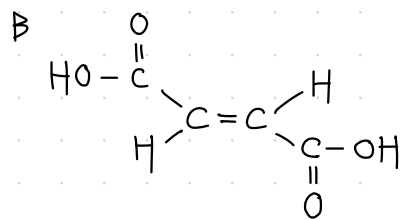
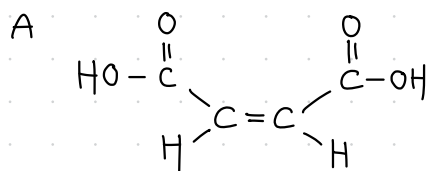
エステルの加水分解生酸性物として $-OH$ 基を持つものがない
 $\rightarrow$ エノール型 か ケト型 に変わった。



問3



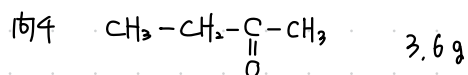
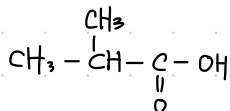
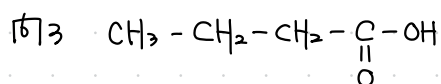
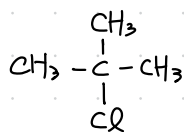
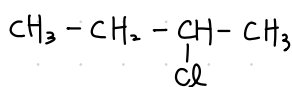
3-8



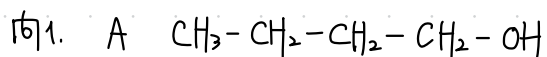
3-9

1 4

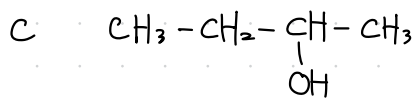
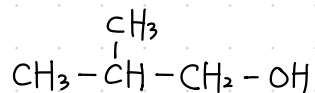
2.



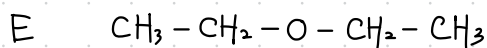
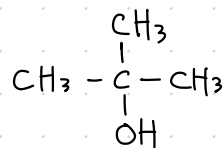
3-10



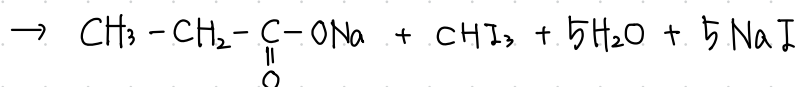
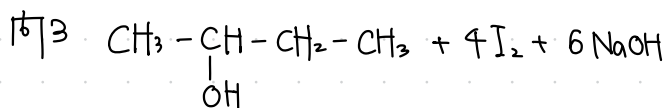
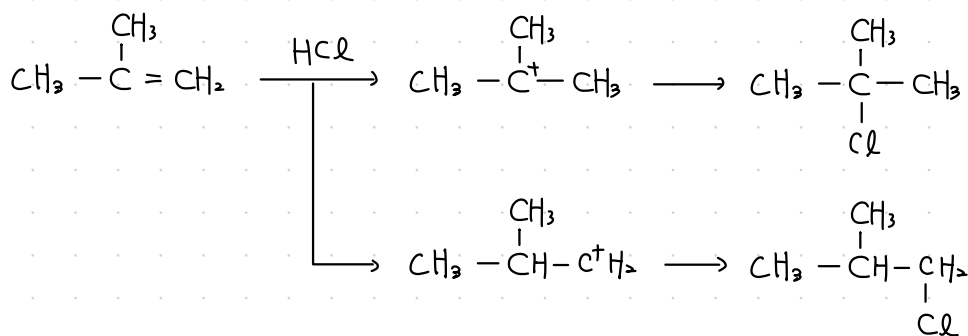
B



D



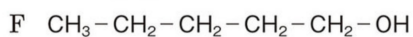
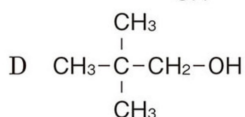
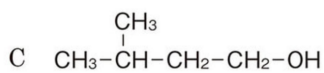
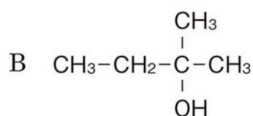
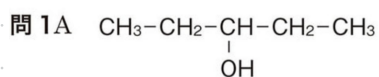
162



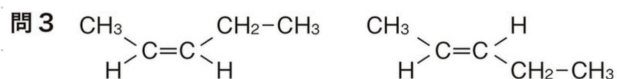
74 C

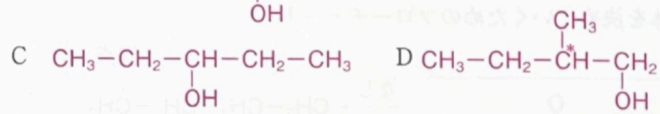
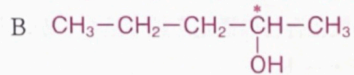
問5. I-TEC 結合

3-11

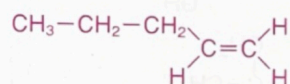
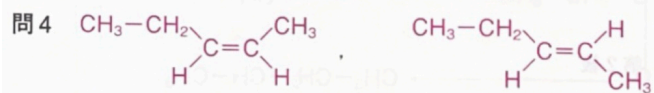


問 2 K

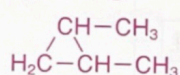
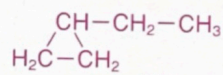
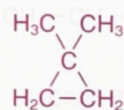
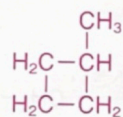
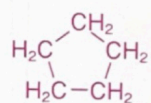


問1 A $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--CH}_2\text{--CH}_2\text{--OH}$ 

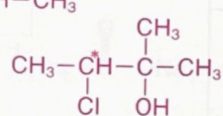
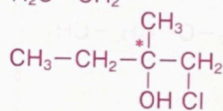
問2 0.15 L 問3 銀鏡反応, アルデヒド基



問5

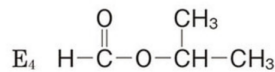
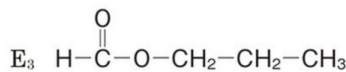
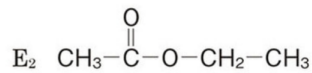
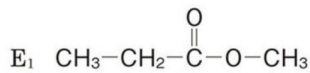


問6

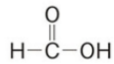


3-13

問1



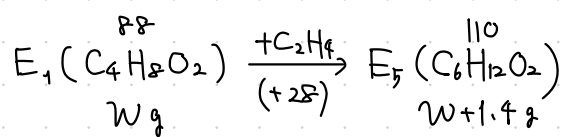
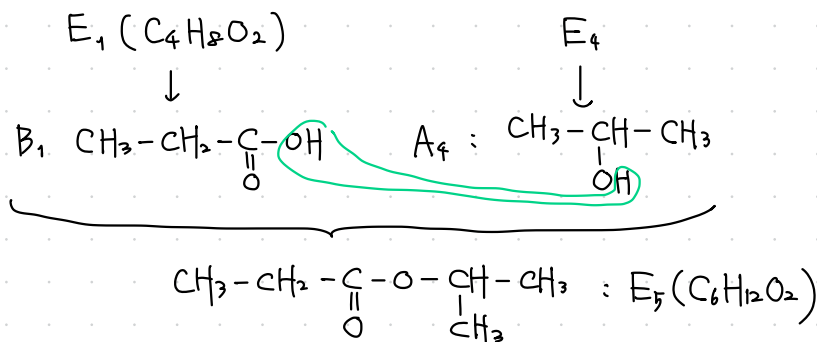
問2 ギ酸は分子内にホルミル基を持つため。(19字)



問3 酢酸カルシウム



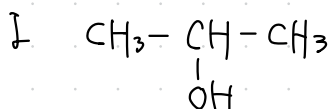
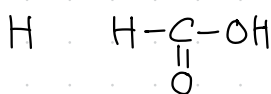
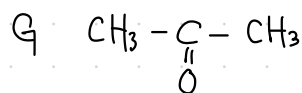
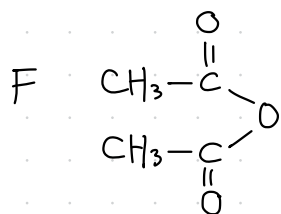
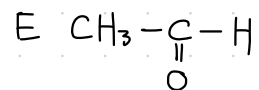
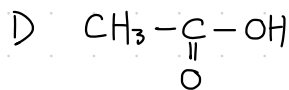
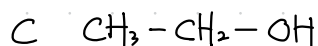
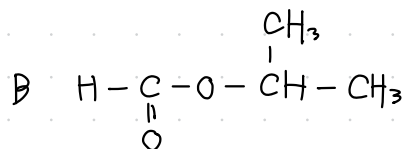
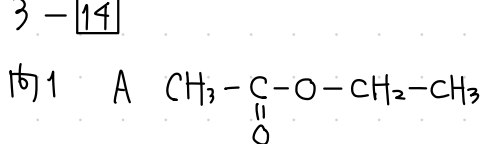
問4 4.4 g



$$\frac{\text{w g}}{88 \text{ g/mol}} = \frac{\text{w} + 1.4 \text{ g}}{110 \text{ g/mol}}$$

$$\therefore \text{w} = \underline{4.4 \text{ g}}$$

3-14



問2.3 略

問4. テトラアルデヒド

問5. ジエチルエーテル

